
METODE INTELIGENTE DE REZOLVARE A PROBLEMELOR REALE

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Graph-based learning

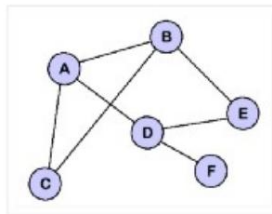
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Universitatea Babeş-Bolyai

Graph neural networks

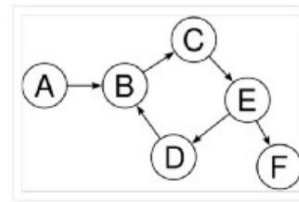
- Why?
- How?
- Applications

Graphs

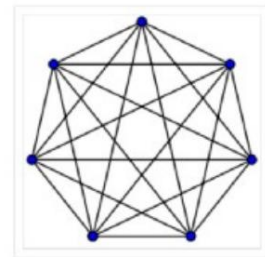
- a type of data structure having two components:
 - nodes (or vertices)
 - Homogenous
 - Heterogenous
 - edges, which connect two nodes
 - Unidirectional
 - Bidirectional
 - With or without weights
- a graph = a collection of loosely inter-connected nodes via edges



undirected graph



directed graph



complete graph

Graph NNs

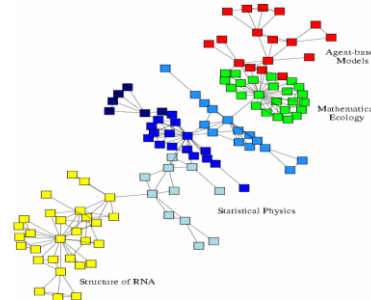
- ❑ neural network architectures that operate on a graph.
- ❑ Aim:
 - for each node in the graph to learn an embedding containing information about its neighborhood (nodes directly connected to the target node via edges).
 - This embedding can then be used for different problems like node labelling, node prediction, edge prediction, etc.
- ❑ Real-life applications
 - Social Network Analysis — Similar posts prediction, tags prediction, and recommending content to users.
 - Natural Sciences — GNNs have also gained popularity in dealing with molecular interactions like protein-protein interactions.
 - Recommender Systems — A heterogenous graph can be used to capture relationships between users and items to recommend relevant items to a buyer.

Why graphs?

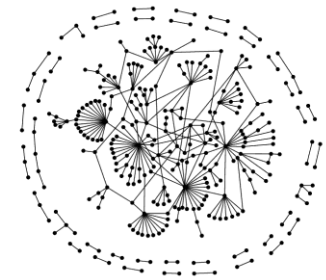
- Graphs (= networks, systems) are a general language for describing and modeling complex systems
- Many data are represented as graphs



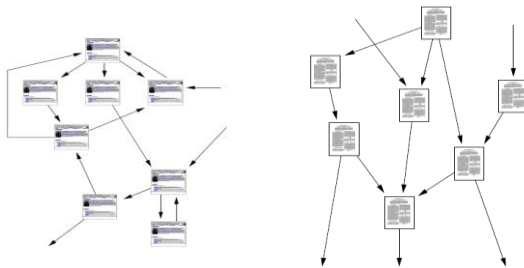
Social networks



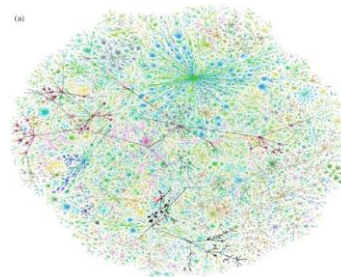
Economic networks



Biomedical networks

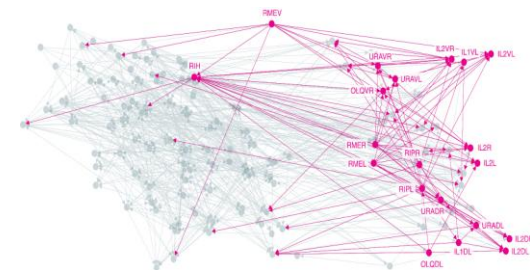


Information networks:
Web & citations



Internet

5



Networks of neurons

Why graphs?

- ❑ Graphs (= networks, systems) are a general language for describing and modeling complex systems
- ❑ Many data are represented as graphs
- ❑ Universal language for describing complex data
 - Networks from science, nature, and technology are more similar than one would expect
- ❑ Shared vocabulary between fields
 - Computer Science, Social science, Physics, Economics, Statistics, Biology
- ❑ Data availability (+computational challenges)
 - Web/mobile, bio, health, and medical
- ❑ Impact!
 - Social networking, Social media, Drug design

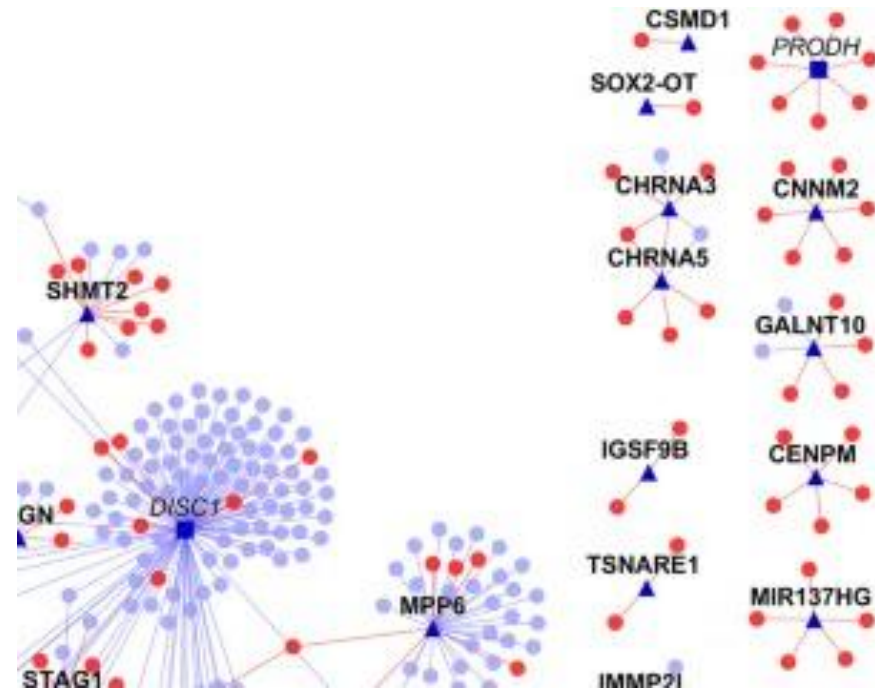
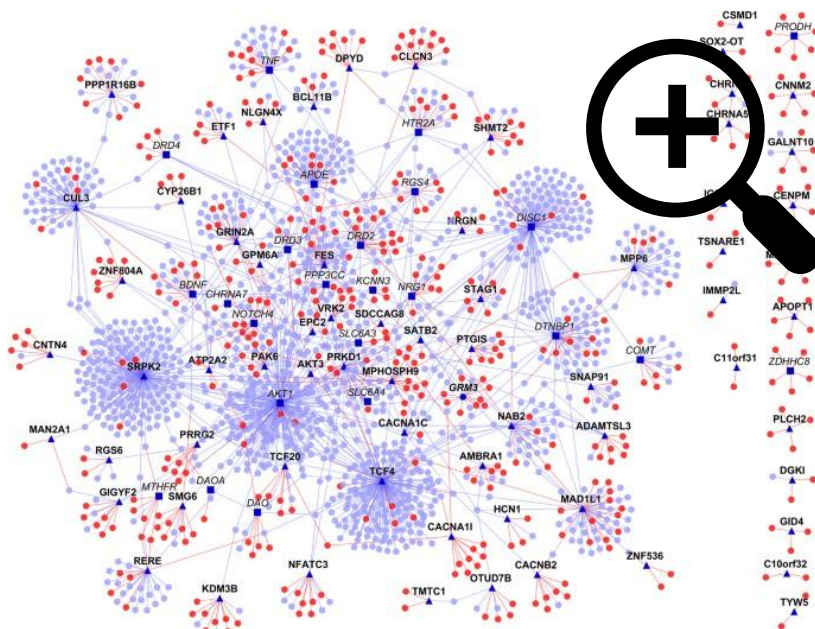
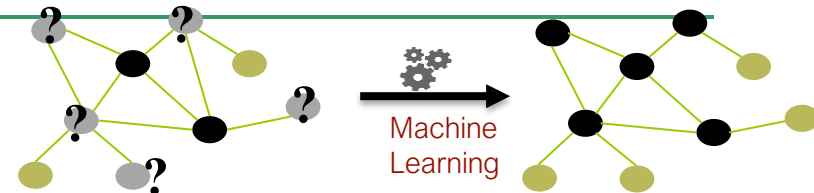
Machine learning with / in graphs

□ Node classification

- Predict the type of a given node

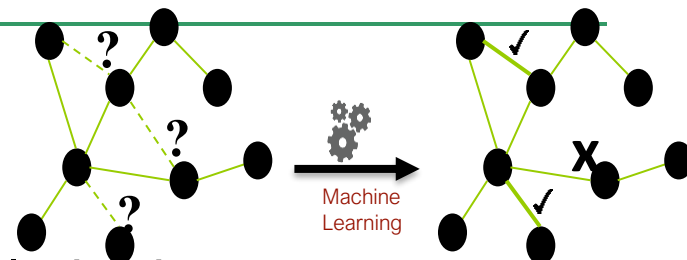
- Classifying the function of proteins in the interactome!

- See Ganapathiraju, M. K., Thahir, M., Handen, A., Sarkar, S. N., Sweet, R. A., Nimgaonkar, V. L., ... & Chaparala, S. (2016). Schizophrenia interactome with 504 novel protein–protein interactions. *NPJ schizophrenia*, 2(1), 1-10. <https://www.nature.com/articles/npjSchz201612>



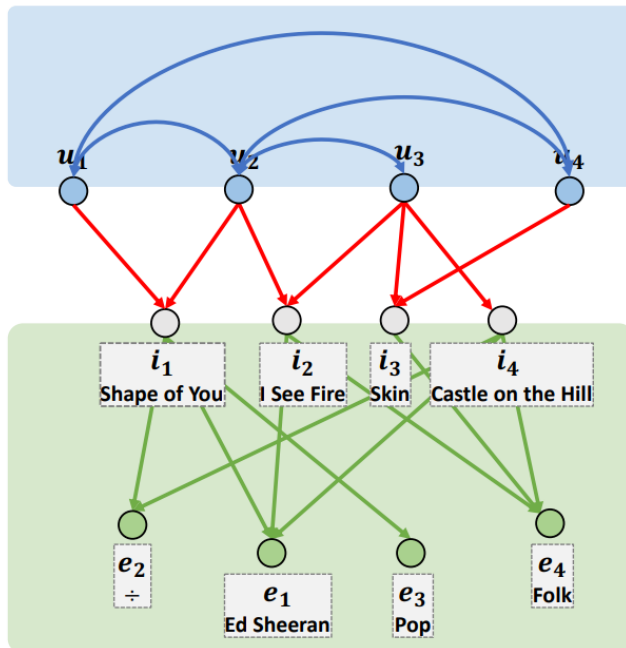
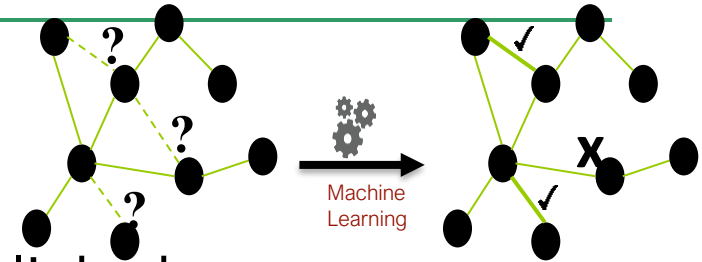
Machine learning with / in graphs

- Node classification
- Link prediction
 - Predict whether two nodes are linked
 - ?? app ??



Machine learning with / in graphs

- Node classification
- Link prediction
 - Predict whether two nodes are linked
 - Recommender systems



User-User Connections

- Social Relations
- Same Profiles ...

User-Item Interactions

- Implicit Feedback
- Explicit Feedback ...

Item-Item Connections

- Same Attributes
- External Knowledge ...



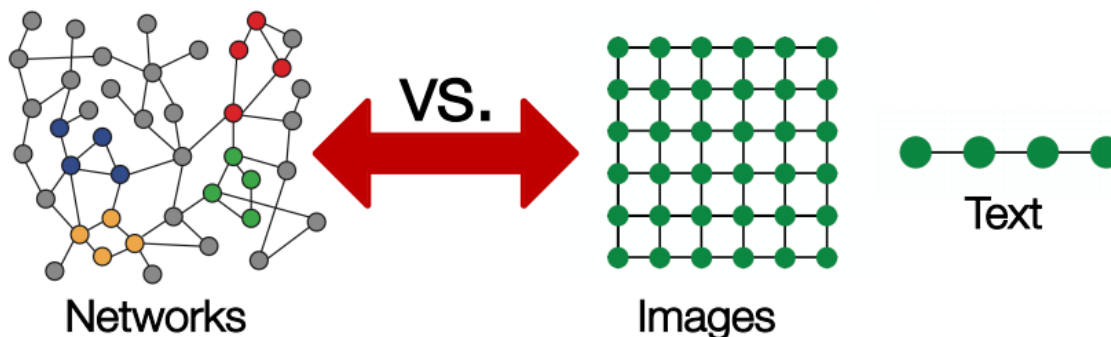
Machine learning with / in graphs

- Node classification
 - Predict a type of a given node
- Link prediction
 - Predict whether two nodes are linked
- Community detection
 - Identify densely linked clusters of nodes
- Network similarity
 - How similar are two (sub)networks

Graph neural networks

□ Data structures

- Text/speech → regular sequences → RNN
- Images → regular grids (matrix) → CNN
- Graphs
 - Arbitrary size
 - Complex topological structure



1. Arbitrary Structure & Size:
Unlike a 256x256 pixel image, graphs have no fixed size or regular topology.

2. No Fixed Node Ordering:
There is no “top-left” node. A GNN must be invariant to the order in which nodes are presented.

3. Variable Neighbourhoods:
Some nodes have two neighbours, others have two thousand. The model must handle this dynamic connectivity.

Graph neural networks

□ Simple ANNs

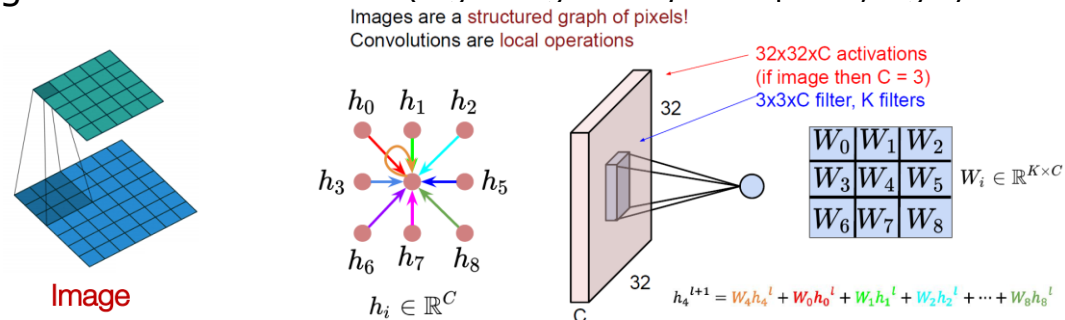
- A simple ANN fed an adjacency matrix is not a viable solution
 - It is not scalable
 - Can not handle graph of different sizes
 - Is not invariant to node ordering

□ Convolutional NNs

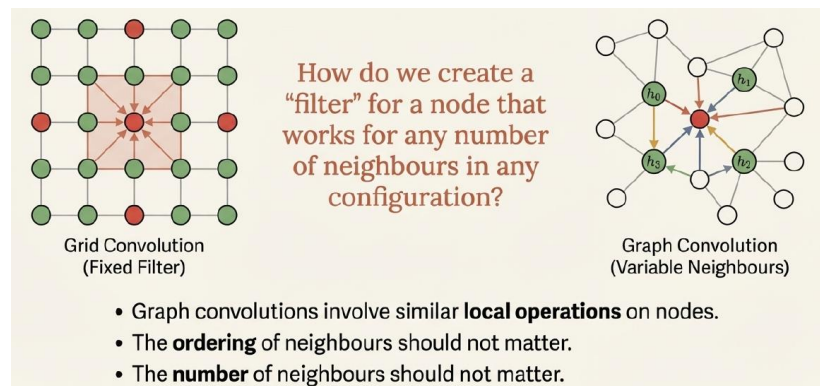
Graph neural networks

□ Convolutions over data

- Combine the information from the current element with that from neighbours $\sum w_i h_i$
 $\rightarrow$ local aggregation of information (e.g. Images are just a special, highly structured type of graph)



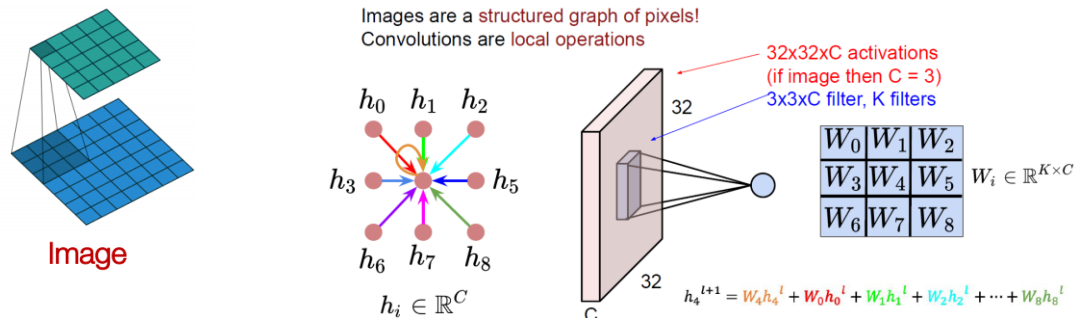
- How to deal with more complex graphs?



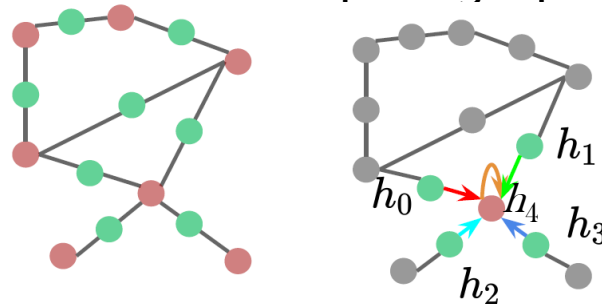
Graph neural networks

□ Convolutions over data

- Combine the information from the current element with that from neighbours $\sum w_i h_i \rightarrow$ local aggregation of information



- How to deal with more complex graphs?



- Graph convolutions involve similar **local operations** on nodes.
- Nodes are now object representations and not activations
- The **ordering of neighbors** should not matter.
- The **number of neighbors** should not matter.
- $N(i)$ are the neighbors of node i
- c_{ij} is a normalization constant

$$h_4^{l+1} = W_4 h_4^l + W_0 h_0^l + W_1 h_1^l + W_2 h_2^l + W_3 h_3^l$$

$$h_i^{l+1} = W_i h_i^l + \sum_{j \in N(i)} \frac{1}{c_{ij}} W_j h_j^l$$

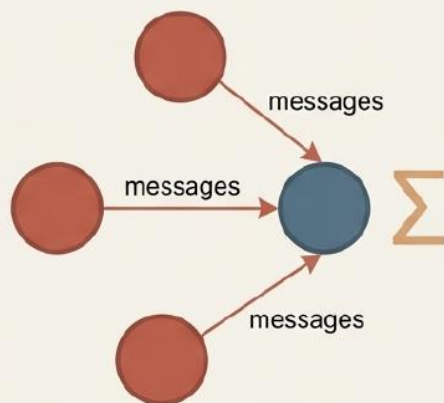
Graph neural networks

□ Convolutions over data

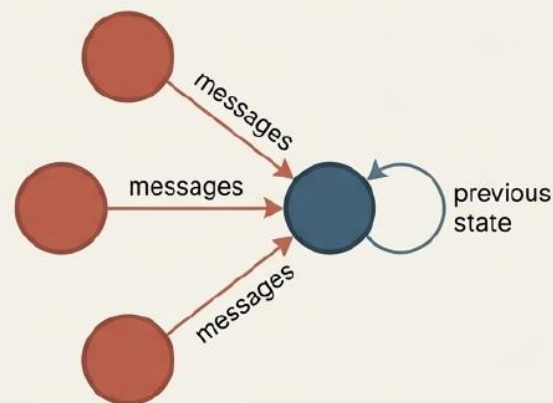
The Core Idea: Neighbourhood Aggregation

GNNs solve this by treating it as a two-step “**message passing**” process. Each node learns about its local environment by iteratively “listening” to its neighbours.

Step 1: AGGREGATE



Step 2: UPDATE

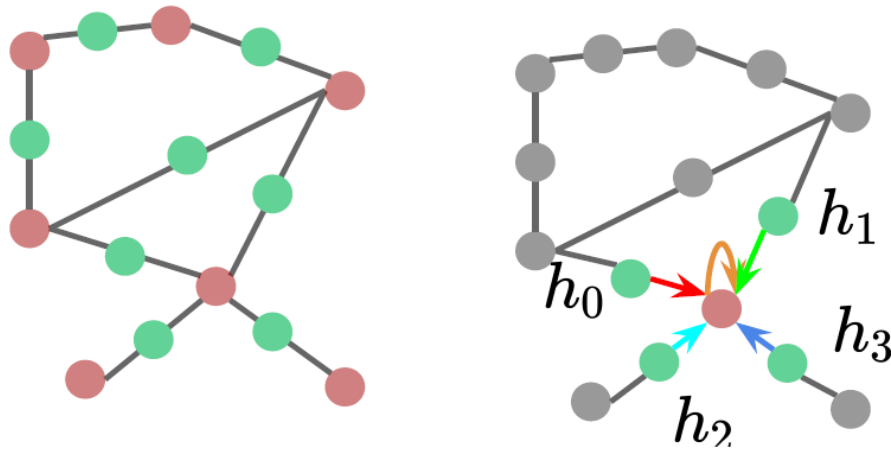


At each layer, a node's representation is updated by aggregating information from its direct neighbours and combining it with its own representation from the previous layer.

$$h_i^{l+1} = w_i h_i^l + \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} w_j h_j^l$$

Graph neural networks

□ Convolutions over graphs



- Updates from some neighbors can be more important than others.
- Attention over neighbors allows graph convolutions to focus on specific neighbors
- σ is a non-linearity, usually ReLU or LeakyReLU.

Without attention:
$$h_i^{l+1} = W_i h_i^l + \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} W_j h_j^l$$

With attention:
$$h_i^{l+1} = W_i h_i^l + \sum_{j \in \mathcal{N}(i)} \frac{1}{c_{ij}} \alpha_{ij} W_j h_j^l$$

where
$$\alpha_{ij} = \frac{e^{\sigma(a^T [Wh_i || Wh_j])}}{\sum_{k \in \mathcal{N}(i)} e^{\sigma(a^T [Wh_i || Wh_k])}}$$

Graph neural networks

□ Naïve approach

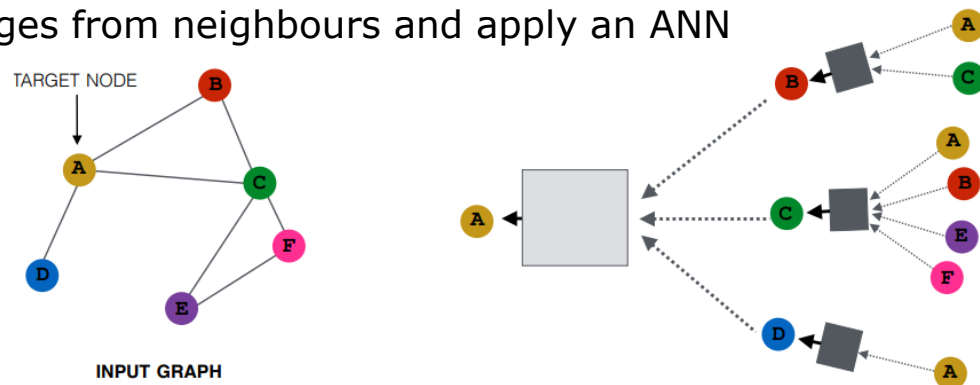
- A fusion between the adjacency matrix and other node features -> input for an ANN
- + easy
- - a lot of parameters = $O(\text{no of nodes})$
- - doesn't work for graphs of different sizes
- - not invariant to node ordering

□ Graph convolutional networks

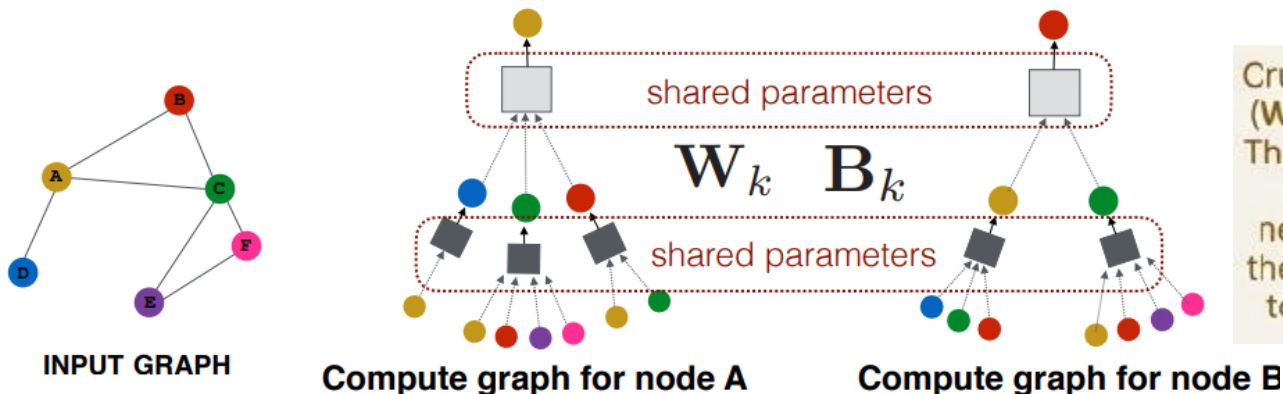
- Node's neighbourhood defines a computational graph
- An ML algorithm is used to learn how to transform and transmit the information across the nodes

Graph convolutional networks

- step1: generate node embeddings based on local graph neighbourhood
 - Aggregate messages from neighbours and apply an ANN



- Every node defines a computation graph based on its neighbourhood

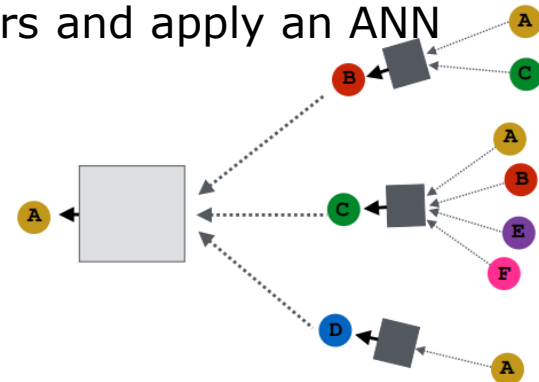
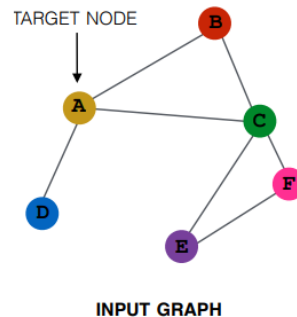


Crucially, the aggregation parameters (W_k, B_k) are **shared** across all nodes. The network learns a single, universal function for how to aggregate neighbourhood information, making the model efficient and generalisable to new nodes or even new graphs.

Graph convolutional networks

- step1: setup for generating node embeddings based on local graph neighbourhood

- Aggregate messages from neighbours and apply an ANN



- Aggregation by an order invariant operator
 - Average (mean) or Max or Sum
 - ANN

Initial 0-th layer embeddings are equal to node features

$$\mathbf{h}_v^0 = \mathbf{x}_v$$

Node v's embedding at layer k

$$\mathbf{h}_v^k = \sigma \left(\mathbf{W}_k \sum_{u \in N(v)} \frac{\mathbf{h}_u^{k-1}}{|N(v)|} + \mathbf{B}_k \mathbf{h}_v^{k-1} \right), \forall k \in \{1, \dots, K\}$$

Learnable weights for neighbours (layer k)

Learnable weights for self (layer k)

Embedding after K layers of neighborhood aggregation

Non-linearity (e.g., ReLU)

Average of neighbor's previous layer embeddings

Previous layer embedding of v

Graph convolutional networks



The GNN acts as a powerful **Encoder**. It takes the graph structure and node features as input and produces a rich numerical vector—a 'node embedding'—for each node.

- step2: prepare the model training
 - Parameters of the graph model: W_k , B_k
 - Classification weights θ
 - Loss function
 - E.g supervised binary classification task = safe or toxic drug

$$\mathcal{L} = \sum_{v \in V} y_v \log(\sigma(\mathbf{z}_v^T \boldsymbol{\theta})) + (1 - y_v) \log(1 - \sigma(\mathbf{z}_v^T \boldsymbol{\theta}))$$

Encoder output: node embedding

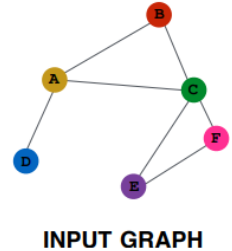
Classification weights

Node class label

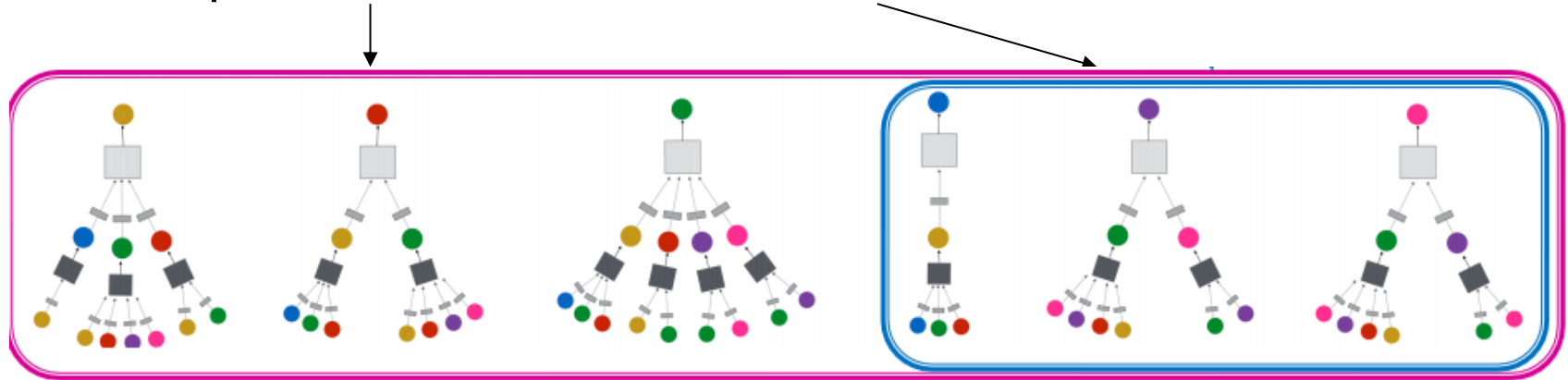
Safe or toxic drug?

- E.g. unsupervised task – random walk optimization
 - See DeepWalk <https://arxiv.org/pdf/1403.6652.pdf>
 - See node2vec <https://cs.stanford.edu/~jure/pubs/node2vec-kdd16.pdf>

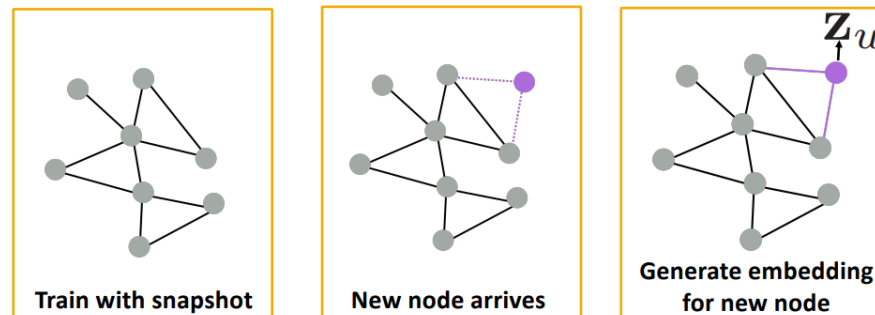
Graph convolutional networks



- step3: train the model and test



- Step 4: generalize for new nodes / graphs



Graph CNNs - applications

■ RecSys

□ Pinterest - PinSage

The Problem

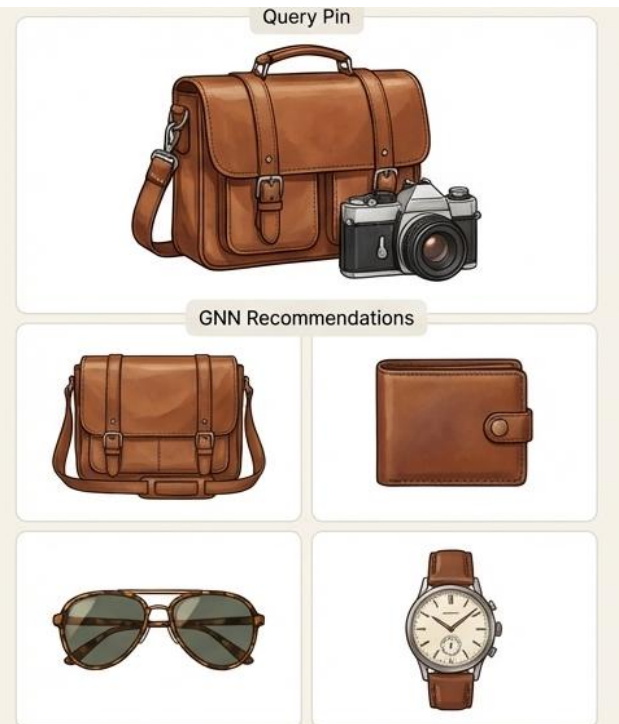
How do you recommend the right 'pin' to a user from a universe of billions of items, especially for new content with no interaction history?

The GNN Solution

Pinterest models its ecosystem as a bipartite graph of over 3 billion nodes (pins and boards) and 18 billion edges. PinSage, a GCN, learns embeddings for every pin based on its visual features and its place in the graph.

The Impact

This allows Pinterest to recommend visually and conceptually similar content, creating a powerful content discovery engine.



Graph CNNs - applications

- RecSys

- Pinterest - PinSage

- <https://medium.com/pinterest-engineering/pinsage-a-new-graph-convolutional-neural-network-for-web-scale-recommender-systems-88795a107f48>
 - <https://arxiv.org/pdf/1806.01973.pdf>

- Model and predict side effects of drug pairs

- <http://snap.stanford.edu/decagon/>
 - <https://arxiv.org/pdf/1802.00543.pdf>

Graph CNNs - applications

■ Data generation

- Drug discovery = Molecule generation (with high value of a given chemical property)

Focus Area: Drug Discovery and Molecular Biology

GNNs are revolutionising how we understand complex biological systems. By representing molecules or protein interactions as graphs, we can predict their properties and relationships.

Predicting Protein Function:

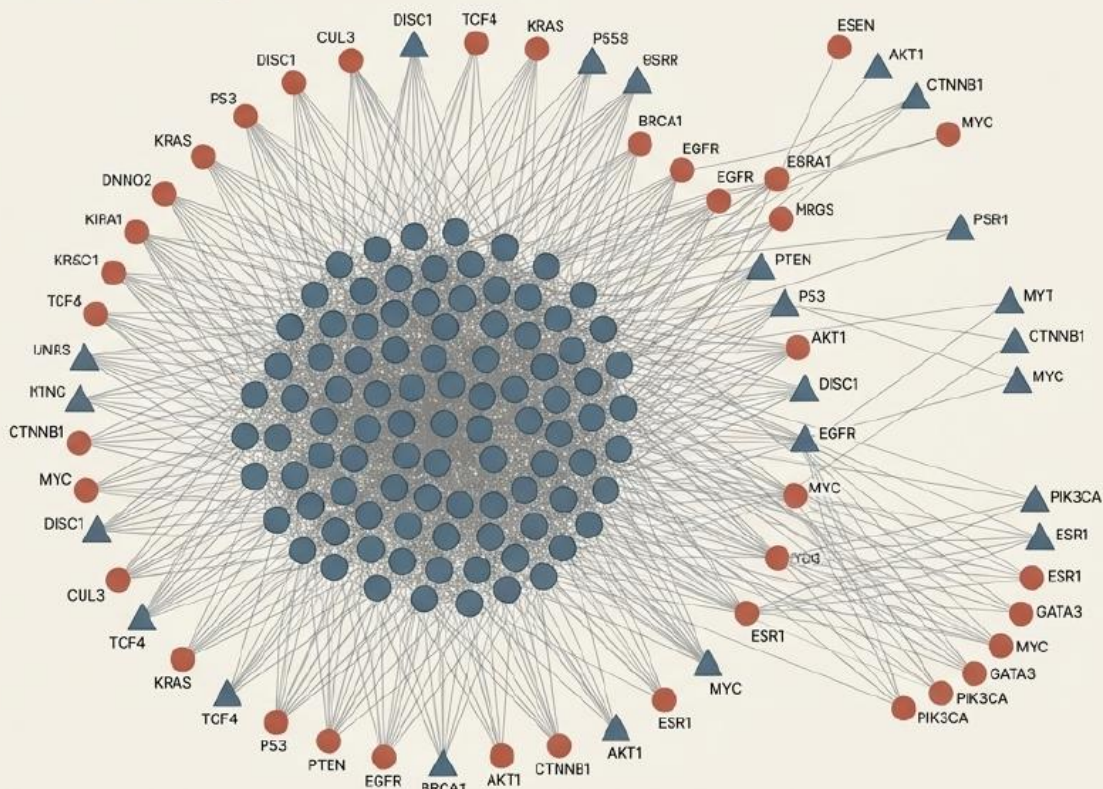
Classifying the function of proteins within the human interactome.

Predicting Drug Side Effects:

The Decagon model predicts the side effects of drug pairs by analysing a massive graph of protein-protein interactions and drug-protein target data.

Molecule Generation:

GNNs can even be used to generate novel molecular structures with desired chemical properties, accelerating drug discovery.



Graph CNNs - applications

- RecSys

- Pinterest - PinSage

- <https://medium.com/pinterest-engineering/pinsage-a-new-graph-convolutional-neural-network-for-web-scale-recommender-systems-88795a107f48>
 - <https://arxiv.org/pdf/1806.01973.pdf>

- Model and predict side effects of drug pairs

- <http://snap.stanford.edu/decagon/>
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- Data generation

- Drug discovery = Molecule generation (with high value of a given chemical property)

- <https://cs.stanford.edu/people/jure/pubs/gcpn-neurips18.pdf>
 - https://github.com/bowenliu16/rl_graph_generation
 - [https://www.cell.com/cell/pdf/S0092-8674\(20\)30102-1.pdf](https://www.cell.com/cell/pdf/S0092-8674(20)30102-1.pdf)

- Action Genome: Understanding Action with Spatio-Temporal Scene Graphs

- <http://actiongenome.org>

- Ji, Krishna et al. Action Genome: Actions as Compositions of Spatio-Temporal Scene Graphs, CVPR 2020

Graph NNs

A New Toolkit for Machine Learning on Graphs

GNNs enable us to perform a range of predictive tasks that were previously intractable on unstructured graph data.



Node Classification

What type is this node?

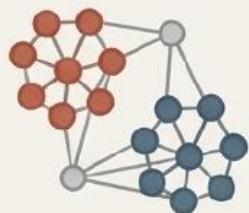
Classifying the function of a protein in a biological network.



Link Prediction

Are these two nodes likely to connect?

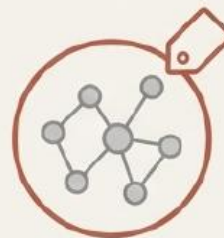
Recommending a new friend on a social network or a product to a user.



Community Detection

What are the densely connected clusters in this network?

Identifying social communities or groups of proteins with related functions.



Graph Classification

What is the property of the entire graph?

Predicting if a molecule is toxic or has a desired chemical property.

Graph NNs

□ Additional information

■ Code example for GNN

- <https://colab.research.google.com/drive/1DIQm9rOx2mT1bZETeEeVUThxcrP1RKqAn>

■ Data

- SNAP project
 - <http://snap.stanford.edu/>
- Open Graph Benchmark
 - <https://ogb.stanford.edu/>

■ GNN and RecSys

- <https://github.com/yazdotai/graph-networks#tensorflow-implementations>
- <https://next-nus.github.io/slides/tuto-cikm2019-public.pdf>

■ Graph-based Deep Learning

- <https://github.com/naganandy/graph-based-deep-learning-literature>
- https://www.cs.mcgill.ca/~wlh/grl_book/
- <https://github.com/thunlp/GNNPapers>

□ Materials are considered from various sources like:

■ Fei-Fei Li's lecture about Graph Convolutions

http://vision.stanford.edu/teaching/cs231n/slides/2020/lecture_18.pdf

■ Jure Leskovec's Lecture about Graph NNs <https://web.stanford.edu/class/cs224w/>

■ <https://www.pyg.org/>

■ Graph Representation Learning Book, by W. Hamilton https://www.cs.mcgill.ca/~wlh/grl_book/

■ Network Science, by A. Barabasi <https://networksciencebook.com/>

■ ...

Beyond the Grid: Learning from the Language of Connection.

Three Key Takeaways



The Big Idea: GNNs are a powerful evolution in AI that allows us to learn directly from relational data, moving beyond simple sequences and grids.



The Core Mechanism: They work by iteratively passing messages and aggregating information between connected nodes, **learning representations** of their local network structure.



The Impact: They are unlocking new frontiers in science (drug discovery), e-commerce (recommender systems), and social network analysis.